Sprint 4 - checklist

1. Decide which user stories can be addressed with machine learning, develop new ones if required
2. Compare different machine learning algorithms on the dataset shared by the lecturer, and select the best one(s) for the user stories that we want to satisfy
3. Embed the machine learning model(s) selected in your web application, and extend the interface so that the machine learning model is used in a meaningful manner i.e. the user stories associated with machine learning capabilities are satisfied
4. Develop and include one of the additional optional features listed in the assignment

Machine learning:

- Develop a predictive model to estimate bike availability using historical station and weather data
- Engineer relevant features (time-based and environmental factors) to improve model accuracy
- Evaluate and select the best-performing model based on test performance
- Integrate the trained model into the backend to support real-time predictions

Chatbot and AI Integration:

- Design and implemented an AI-powered chatbot for user interaction
- Integrate a third-party AI service to generate dynamic responses
- Develop logic to handle user queries and provide contextual assistance
- Test chatbot behaviour across typical and edge-case scenarios

Authentication System and Login:

- Design and implement a user authentication system with secure login and registration
- Connect frontend interfaces with backend validation and database operations
- Implement session management to maintain user login state
- Add support for third-party authentication providers

Frontend Features and Integration:

- Integrate frontend components with backend APIs to enable dynamic data updates
- Implement core user flows including account creation and navigation between pages
- Ensure consistency across pages through alignment with the design system
- Finalise UI components for authentication and user interaction

Map and Core Features Improvements:

- Enhance map performance and responsiveness across different screen sizes
- Improve station interaction and real-time data display

- Implement and refine route planning functionality
- Ensure consistent integration of weather and station data across the application

UX/UI Polish:
- Refine overall visual design, including colour system and layout consistency
- Improve spacing, alignment, and component structure across pages
- Enhance usability and accessibility of the interface
- Apply final branding elements and visual polish

Testing and Debugging:
- Conduct comprehensive testing of all core features (map, authentication, chatbot, ML, etc.)
- Identify and resolve bugs and edge cases
- Perform cross-browser and responsiveness testing
- Validate system behaviour under different user scenarios

Final Refactoring:
- Improve code structure, readability, and modularity
- Standardise naming conventions across the codebase
- Add comments and documentation for maintainability
- Remove redundant or unused code and assets

Project Report:
- Complete all report sections including system overview, design, and implementation
- Document machine learning approach and evaluation
- Summarise testing strategy and results
- Produce sprint review, retrospective, and burndown analysis
- Finalise report formatting, proofreading, and future work discussion